

Executive Summary — Customer Churn Analysis

Business Overview

The analysis examines customer churn across **7,043 telecom customers and 21 attributes**, covering customer demographics, tenure, services, contracts, payment methods, and charges. The objective is to identify the major customer segments associated with churn and highlight areas where retention efforts can be focused.

Key Findings

- **Overall churn is significant:** The analysis identifies a **26.54% customer churn rate**, meaning more than one in four customers have left the service.
- **Senior citizens show a higher churn tendency:** The analysis indicates that a greater percentage of senior-citizen customers have churned compared with non-senior customers.
- **Tenure is strongly associated with churn:** Customers with **very short tenure**, particularly those who have used the service for only around **1–2 months**, are more likely to churn, while customers with longer tenure tend to remain.
- **Contract type is a major churn indicator:** Customers on **month-to-month contracts** are considerably more likely to churn than customers with one-year or two-year contracts.
- **Payment method matters:** The analysis highlights **electronic-check customers** as a segment with a higher likelihood of churn.
- **Gender does not appear to be a primary driver:** The notebook compares churn across male and female customers, but the analysis does not identify gender as a major churn factor.

Customer Profile at Risk

The analysis suggests that the highest-priority churn-risk segment consists of customers who have:

Short tenure + Month-to-month contract + Electronic check payment

with **senior-citizen customers** representing another segment requiring additional attention. These patterns point toward customer retention being particularly important during the **early stages of the customer lifecycle**.

Recommended Business Actions

1. **Focus on the first few months**
 - Introduce onboarding programs and early-stage customer engagement.

- Monitor customers during their first 1–3 months.
- 2. **Encourage longer-term contracts**
 - Provide incentives for customers to move from month-to-month contracts to one- or two-year plans.
- 3. **Target electronic-check customers**
 - Encourage alternative payment methods such as automatic bank transfer or credit card.
 - Investigate whether payment experience is contributing to churn.
- 4. **Develop targeted senior-citizen retention programs**
 - Provide additional support and personalized service for this segment.
- 5. **Use churn-risk segmentation**
 - Instead of treating all customers equally, prioritize customers combining multiple risk factors such as short tenure, month-to-month contracts, and electronic-check payments.

Executive Takeaway

Customer churn is primarily concentrated among customers with shorter relationships and less-committed service arrangements. Month-to-month customers, particularly those in the early stages of their tenure and those using electronic checks, represent the most important retention opportunity. Senior citizens also show elevated churn behavior. The business should therefore prioritize early customer engagement, incentives for longer-term contracts, and targeted retention initiatives for high-risk customer segments.